Family Name ____________________

First Name ____________________

Student Number |__|__|__|__|__|__|__|__|

Venue ____________________

Seat Number ________

UNIVERSITY OF CANTERBURY
Te Whare Wānanga o Waitaha
CHRISTCHURCH NEW ZEALAND

No electronic/communication devices are permitted.

No exam materials may be removed from the exam room.

Electrical and Computer Engineering

EXAMINATION

Mid-Year Examinations, 2024

ENEL445-24S1 (C) Special Topic: Applied Engineering Optimisation

For Examiner Use Only

Question	Mark

Total ________

Examination Duration: 120 minutes

Exam Conditions:

Restricted Book exam: Students may bring in the listed approved materials only.

Any scientific/graphics/basic calculator is permitted.

No exam materials may be removed from the exam room.

Materials Permitted in the Exam Venue:

Restricted Book exam materials:

One A4 double-sided page of notes - printed/copied/handwritten.

Materials to be Supplied to Students:

1 x Write-on exam paper

Instructions to Students:

ALL questions are COMPULSORY.

There are FOUR (4) questions in total - of UNEQUAL value.

The maximum total mark is 85.

There is extra space at the end of this answer booklet should you need it.

Show your working to receive partial credit.

Indicate when your work is continued on an extra page.

INSTRUCTIONS

This is an closed-book test (apart from the A4 cheat sheet). Please write your answers in the space provided. There is extra space at the end of the test should you need it. Show your working as you complete the test, so that you can more easily receive partial credit. Make a note indicating when your work is continued on an extra page.

Q1. Gradient-free Optimisation Part I

(a) [3 marks]

Give **three** situations where gradient-free methods are more suited than gradient-based methods for solving an optimisation problem.

(b) [2 marks]

On the objective function contour plot below, carefully sketch the expansion operation for the Nelder-Mead method. No need to continue to complete the iteration.

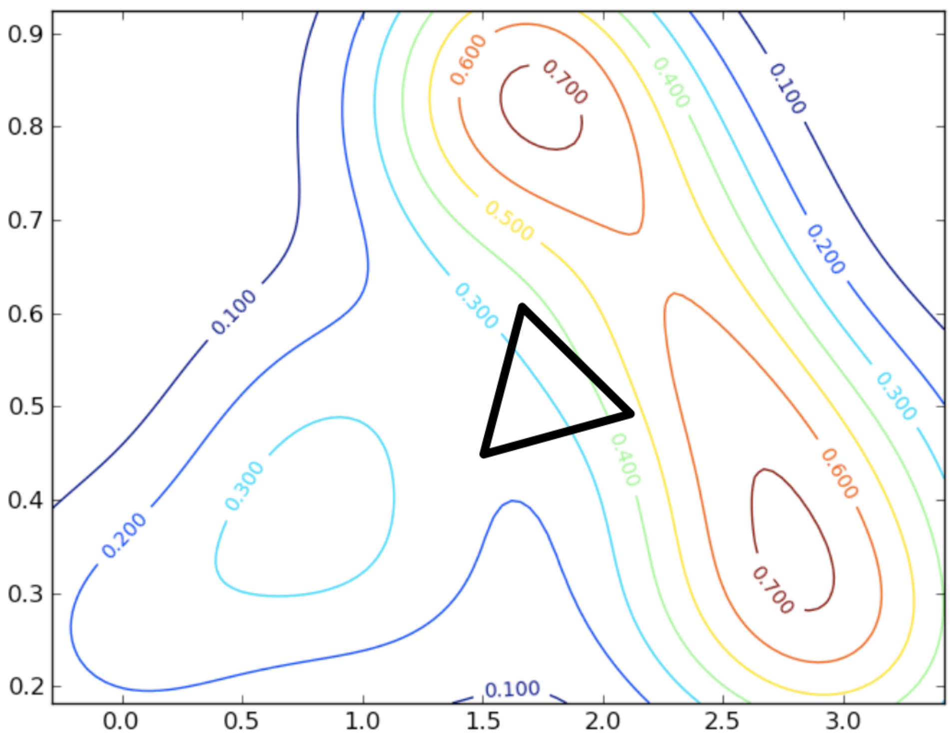

Figure 1: Contour plot for question 1(b)

(c) [3 marks]

Name **three** convergence metrics that can be used with the Nelder-Mead method.

(d) [2 marks]

Sketch a minimal positive spanning set of vectors in 2-dimensions **AND** state the process of opportunistic polling in the context of the generalised pattern search algorithm.

(e) **[3 marks]**

What are the **two** significant shortcomings of Shubert's algorithm and briefly explain how these two shortcomings are addressed by the DIRECT method.

(f) **[3 marks]**

On the f-d plot below, circle the dots representing the rectangles that will be subdivided next by the DIRECT method. The value for $\epsilon|f_{\min}| = 3$. Assume all point-shifts have been done.

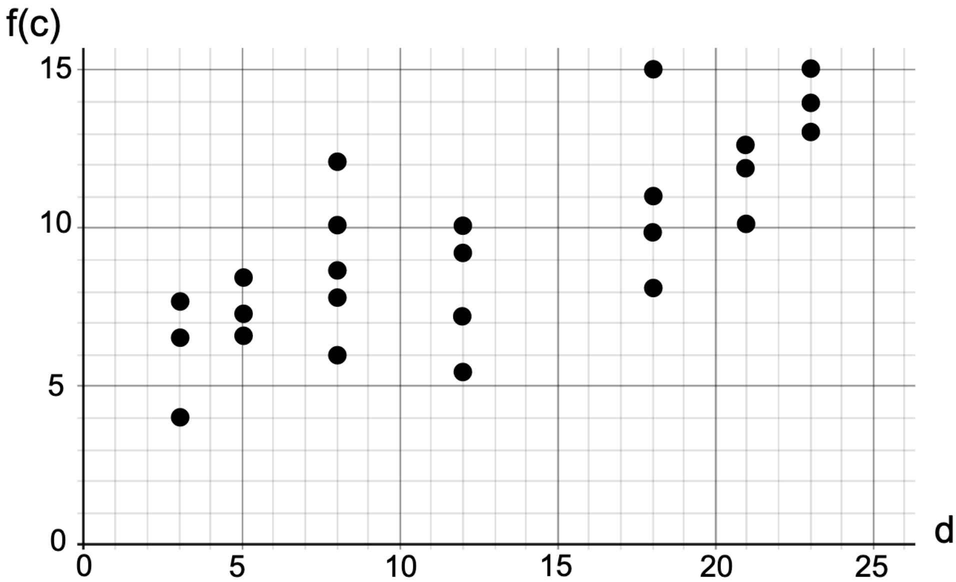

Figure 2: f-d plot for question 1(f)

(g) [4 marks]

Briefly explain the three main steps in a genetic algorithm and discuss the advantages and disadvantages of a binary-coded gene versus a real-coded gene.

Q2. Gradient-free Optimisation Part II

(a) [5 marks]

Convert the following linear programming problem into standard form

$$
\begin{array}{ll}
\text{maximise} & 2x_1 - x_2 + 8x_3 \\
\text{subject to} & 7x_1 + 5x_2 - x_3 \leq 1 \\
 & \left| x_1 + 6x_2 \right| \geq 4 \\
 & x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \text{ free}
\end{array}
$$

(b) [4 marks]

Solve the following optimal assignment problem (find the assignment with the least cost) using the Hungarian algorithm:

	Task 1	Task 2	Task 3
Worker A	100	150	125
Worker B	150	175	135
Worker C	120	250	140

(c) [2 marks]

State the purpose of the relaxation step in the branch-and-bound algorithm **AND** give an example of a relaxation in a mixed-integer programming problem.

(d) [3 marks]

Define the Pareto front and explain how it can be used to gain insights into an optimisation problem.

(e) **[6 marks]**

Given M measurements $\{g_n\}$ and a model $m(\mathbf{c}, y_n)$, where y_n is the nth input, and $\mathbf{c}$ contains parameters of the model, show mathematically that maximising the log likelihood is equivalent to minimising the sum of squared errors, $\{z_n\}$, between the measurement and the model if these errors are distributed as a zero-mean, equal variance Gaussian.

Q3. Unconstrained Optimisation

(a) Consider generating the generalised Van der Corput sequence for initializing unconstrained optimization algorithms.

(i) [2 marks]

Let i be a positive integer. We first need to represent i in base b as

$$i = a_0 + a_1 b + a_2 b^2 + \cdots + a_r b^r, \tag{1}$$

where $0 < a_r \leq b-1$ and $0 \leq a_j \leq b-1$ for $j = 0, 1, 2, ..., r-1$. That is, i can be expressed as a $(r+1)$-digit, $(a_r a_{r-1} \cdots a_0)_b$, in base b.

Find, using (1), the representation of $i = 1, 2, 3, 4$ in base $b = 3$.

(ii) [2 marks]

The ith element in the generalized Van der Corput sequence can then be computed using the following radical inversion function for base b:

$$\phi_i(b) = \frac{a_0}{b} + \frac{a_1}{b^2} + \cdots + \frac{a_r}{b^{r+1}}. \tag{2}$$

Using your results in the previous question and (2), find $\phi_i(3)$ for $i = 1, 2, 3, 4$.

(b) With the Newton's method, at iteration k, the objective function $f(\mathbf{x})$ is first approximated using its second-order Taylor-series expansion at the solution obtained in the $(k-1)$-th iteration, $\mathbf{x}_{k-1}$, via

$$f(\mathbf{x}) \approx f(\mathbf{x}_{k-1}) + \nabla f(\mathbf{x}_{k-1})^T(\mathbf{x} - \mathbf{x}_{k-1}) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_{k-1})^T\mathbf{H}(\mathbf{x}_{k-1})(\mathbf{x} - \mathbf{x}_{k-1}). \qquad (3)$$

Here, $\nabla f(\mathbf{x}_{k-1})$ is the gradient of $f(\mathbf{x})$ at $\mathbf{x}_{k-1}$, and $\mathbf{H}(\mathbf{x}_{k-1})$ is the Hessian matrix of $f(\mathbf{x})$ at $\mathbf{x}_{k-1}$, which is assumed to be **invertible**.

(i) **[3 marks]**

The approximated objective function in (3) can be minimized by taking the partial derivative with respect to $\mathbf{x}$ and setting the result to zero.

Carry out the above minimization process to find the expression for the updated solution $\mathbf{x}_k$.

(ii) **[4 marks]**

Using your result in the previous question to show that if $\mathbf{H}(\mathbf{x}_{k-1})$ is also positive definite, $(\mathbf{x}_k - \mathbf{x}_{k-1})$ lies in the direction of decreasing the objective function $f(\mathbf{x})$ at $\mathbf{x}_{k-1}$.

(iii) [4 marks]

With the quasi-Newton method, the Hessian matrix $\mathbf{H}(\mathbf{x}_k)$ needs to be estimated as well. For this purpose, we approximate the objective function $f(\mathbf{x})$ around $\mathbf{x}_k$ using

$$f(\mathbf{x}) \approx f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^T(\mathbf{x} - \mathbf{x}_k) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_k)^T \tilde{\mathbf{H}}(\mathbf{x}_k)(\mathbf{x} - \mathbf{x}_k), \tag{4}$$

where $\tilde{\mathbf{H}}(\mathbf{x}_k)$ is the estimated Hessian.

We require that the approximation in (4) can recover the gradient of the objective function at $\mathbf{x}_{k-1}$, $\nabla f(\mathbf{x}_{k-1})$. This is called the secant condition.

Write out the secant condition using (4).

Hint: Taking the partial derivative with respect to $\mathbf{x}$ on both sides of (4) and then setting $\mathbf{x}$ to be $\mathbf{x}_{k-1}$ yields the desired result.

(c) [5 marks]

Ridge regression: consider regularizing the linear least square objective function as

$$f(\mathbf{x}) = \frac{1}{2}(\mathbf{y} - \mathbf{A}\mathbf{x})^T(\mathbf{y} - \mathbf{A}\mathbf{x}) + \frac{\lambda}{2}\mathbf{x}^T\mathbf{x}, \tag{5}$$

where $\lambda > 0$ is the regularization parameter.

Find the optimal solution that minimizes the objective function in (5).

Q4. Power systems optimisation & algorithms

(a) [4 marks]

Estimate the computational expense (in terms of time) of the AC optimal power-flow problem for a power system if: a) the standard (linear) DC optimal power-flow problem for the same power system can be solved in 5 seconds; b) the power system has 20 generators, 1000 buses, and 1000 lines; c) the computational scaling of the solver is approximately $O((n + n_g + n_h)^3)$, where n is the number of decision variables, and n_g and n_h are the number of inequality and equality constraints respectively; and d) nonlinearity is responsible for a five-fold slowdown compared to a linear problem of the same dimensions.

Hint: the number of generators is comparatively insignificant and can be ignored when estimating computational expense.

(b) [6 marks]

Suggest an optimisation algorithm to solve the AC optimal power-flow problem in a), and briefly justify your choice. For the algorithm you have chosen, explain its main concepts - preferably with the aid of one or more equations.

(c) [5 marks]

What is the difference between *security-constrained* AC optimal power flow (SC-OPF) and the regular (i.e. not security-constrained) AC optimal power-flow? Why is the full security-constrained AC optimal power-flow problem difficult to solve in practice?

(d) [2 marks]

Briefly explain how security constraints are included in (industry) practice without incurring huge computational costs.

(e) [8 marks]

Consider the three-bus system depicted in Fig. 3. You may assume that each meter is very close to the nearest bus in the figure.

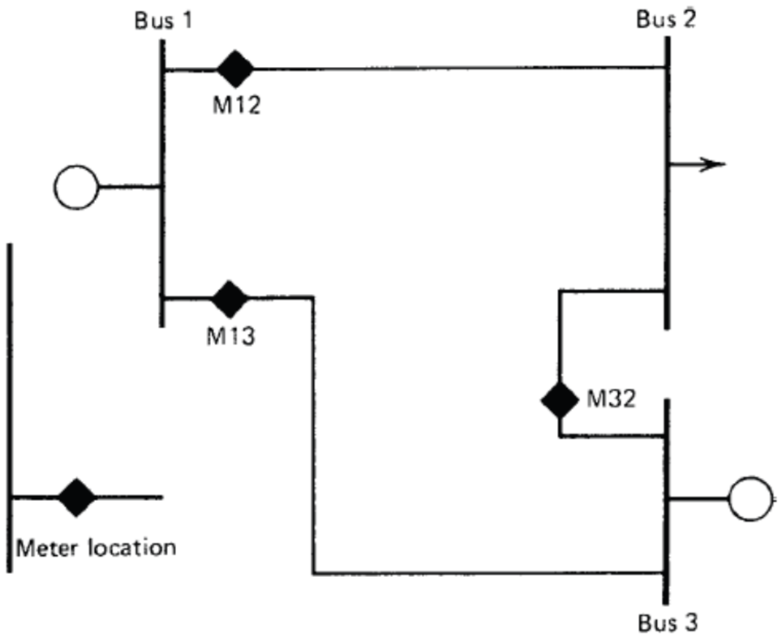

Figure 3: Power system for question 4 (e)

Given (potentially noisy) measurements of $P_{12}, Q_{12}, P_{13}, Q_{13}, P_{32}, Q_{32}$ (i.e. the power-flows of every line) and V_1, V_2 (i.e. the voltage magnitude at buses 1 and 2), **formulate a *non-linear* least squares problem to solve for the power system states (voltage magnitudes and angles at each bus)**. You may assume that all meters have the same standard deviation (in per unit).

The active power flow through the line from i to j is given by:
$P_{ij} = V_i V_j B_{ij} \sin(\theta_i - \theta_j)$.

The reactive power leaving bus i through the line from i to j is given by:
$Q_{ij} = V_i(V_i - V_j \cos(\theta_i - \theta_j))B_{ij}$.

Hint: You do NOT need to solve this problem - simply write the optimisation problem to be solved in a standard form.

Extra Answer Space (Optional)

Extra Answer Space (Optional)

END OF PAPER